

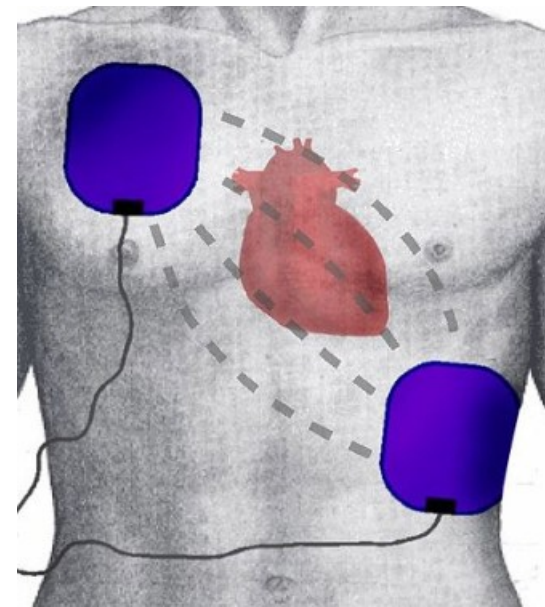
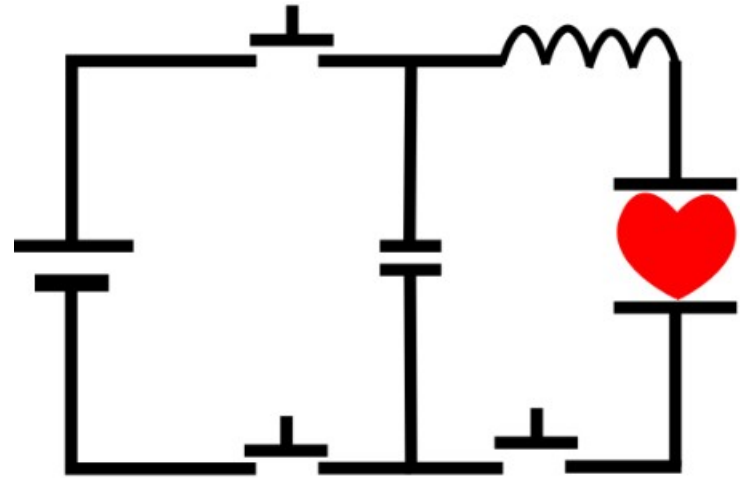
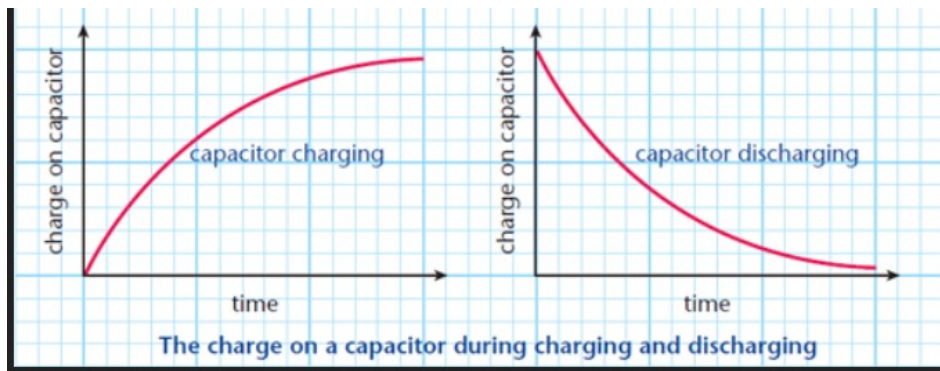
capacitors



<http://micro.magnet.fsu.edu/electromag/java/capacitor/index.html>

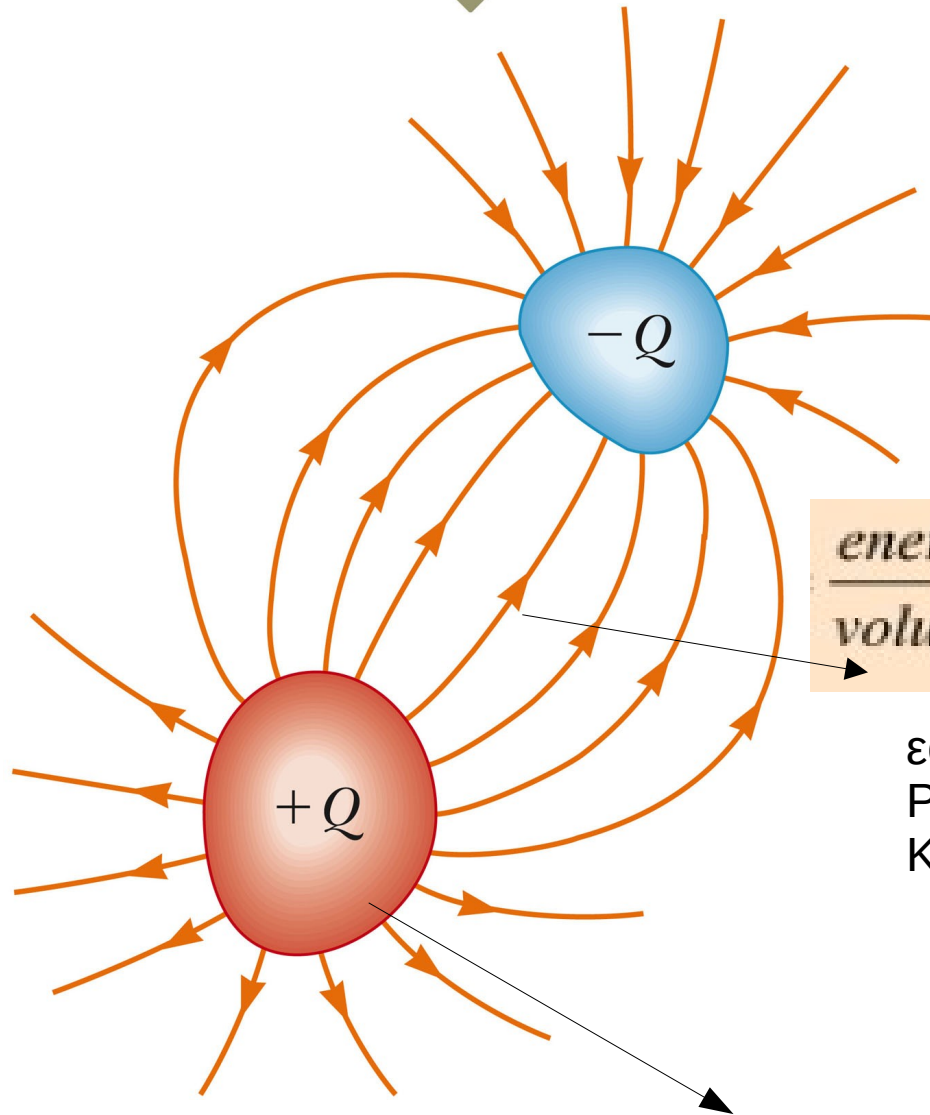
https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html

demonstration



200V → 1000V for heart.

When the capacitor is charged, the conductors carry charges of equal magnitude and opposite sign.



$$\frac{\text{energy}}{\text{volume}} = \frac{1}{2} \epsilon E^2$$

$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$
Permittivity of vacuum
 $K_e = 1 / (4\pi\epsilon_0) = 9 \times 10^9$

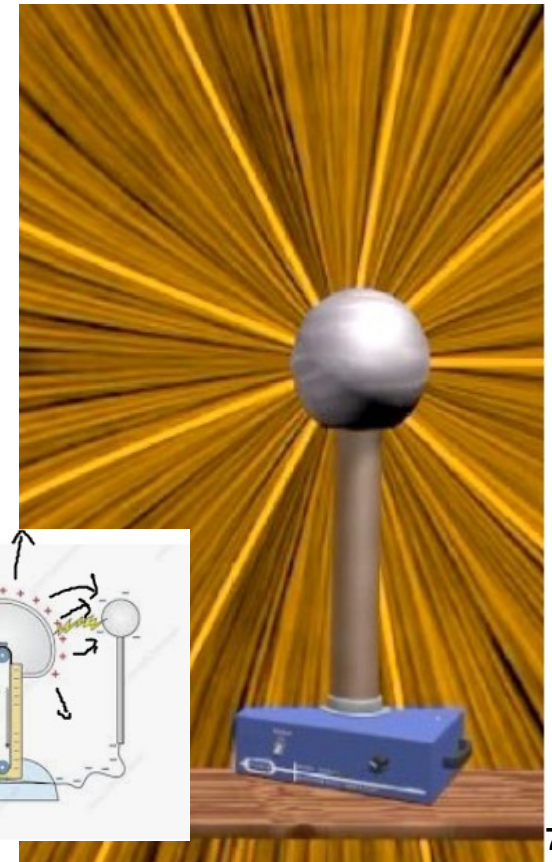
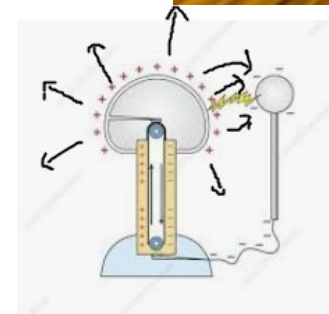
The conductors are Equipotentials
The power supply transfer electrons
From one conductor to the other → work is done

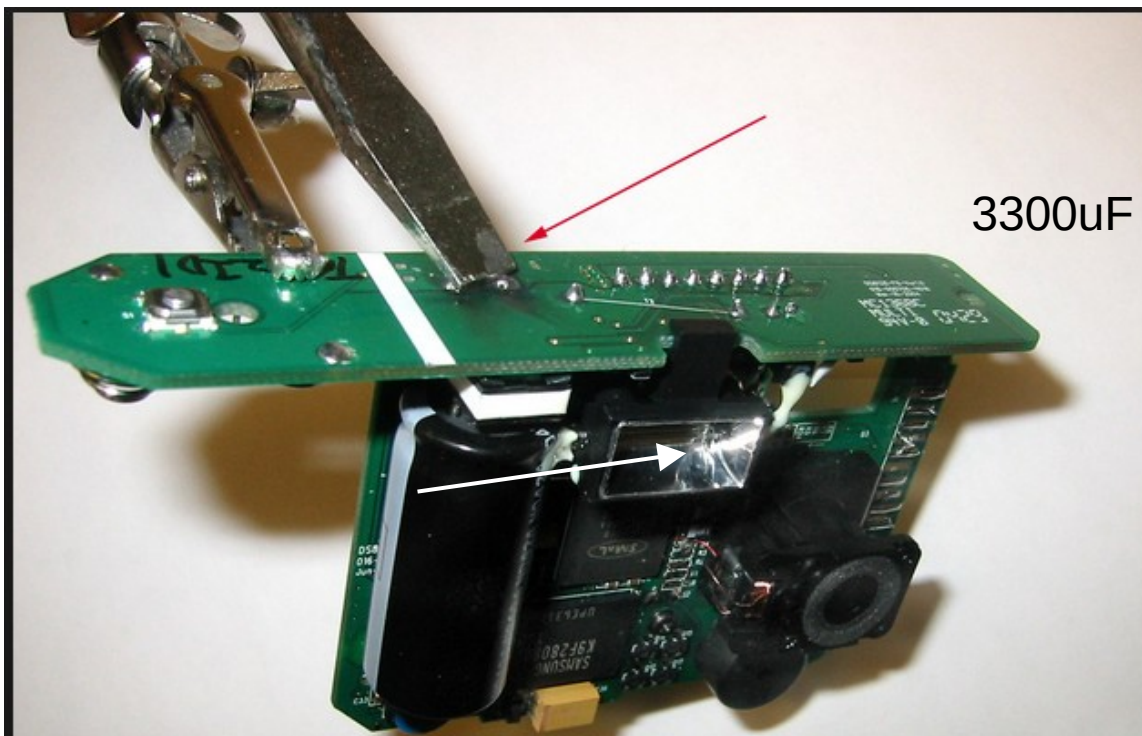
A capacitor = 2 conductors with Opposite charges. The conductors Are called plates.

A insulator (air, plastic ..) is placed Between the conductors.
(called a dielectric)

The electric field takes place between The conductors.

A capacitor is a device to package Energy in an electrostatic field. In a storehouse.





demo

See Harold Edgerton (30s) →

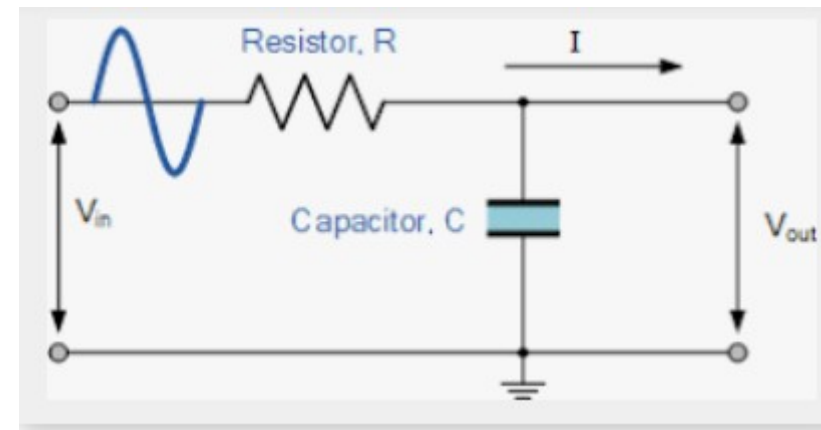
Made high speed photography possible with His flash-tube He used a capacitor That dump the energy in a bulb that contained Xenon gas → plasma discharge (intense Burst of light in μs)
→ strobes and high speed photography

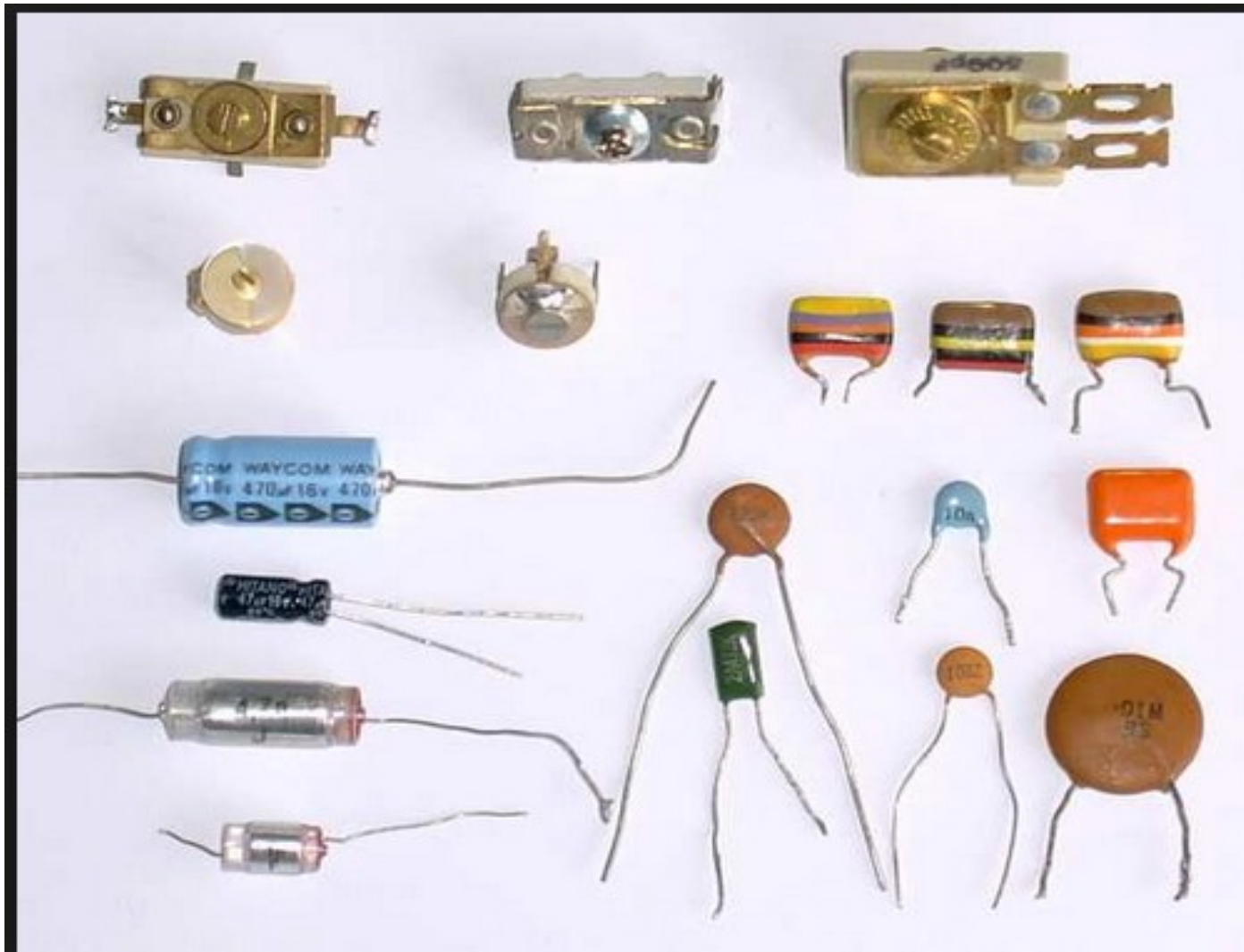


<http://www.ladyada.net/make/sudc4kap/make.html>

- Holds binary numbers (RAM memory)
- make keyboards
- filter noise
- block DC, turn AC to DC
- battery

<https://www.youtube.com/watch?v=EoWMF3Vki6U>
4:43 and 5:23 fun with ultracapacitor





Larger the capacitors, more charge it can hold. The amount of charge it can Hold depends on the geometry. A Van Graaf is spherical and can not hold Has much as those cylinders with rolled conductors inside.
The geometry and therefore the amount of charges → the capacitance.

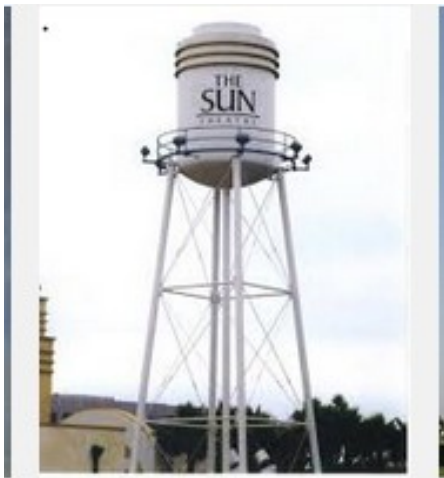
Capacitance → how much charge a cap can hold

The capacitance is a constant.

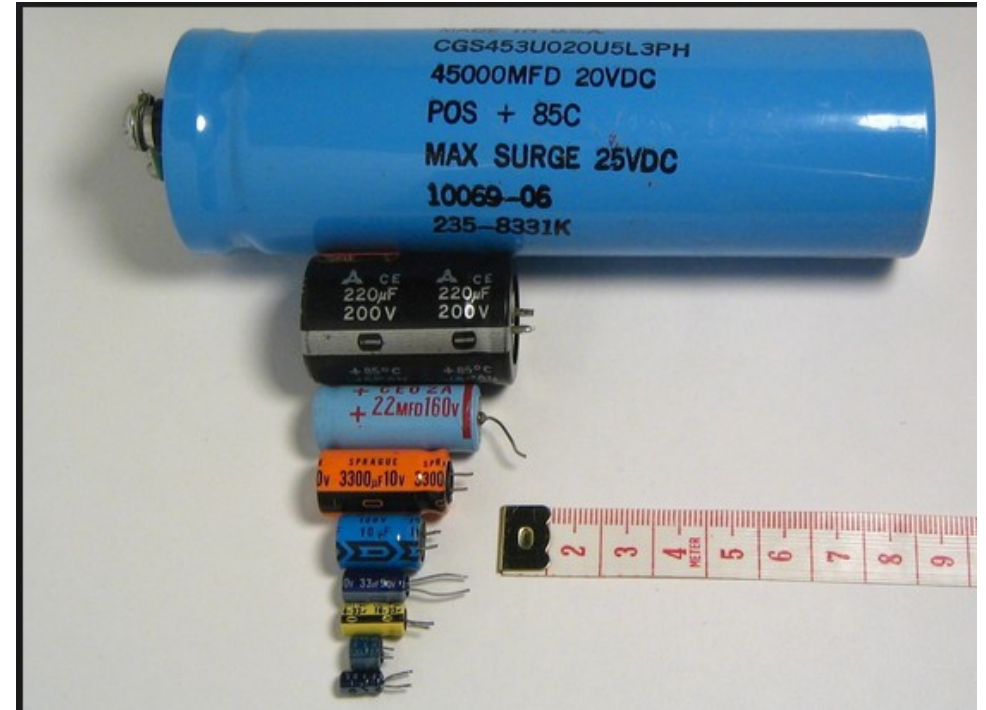
It depends on the geometry of the capacitor and on the dielectric.

It is a measure of how much charge the capacitor can store.

Like the volume of a water tower tells how much water it can hold.



Unit is farad (F). example: 10 microfarad capacitor.



$$C = Q / V$$

Q is in coulomb©

V in volt

C in Farad

SI Unit of Capacitance:
coulomb/volt = farad (F)

THE RELATION BETWEEN CHARGE AND POTENTIAL DIFFERENCE FOR A CAPACITOR

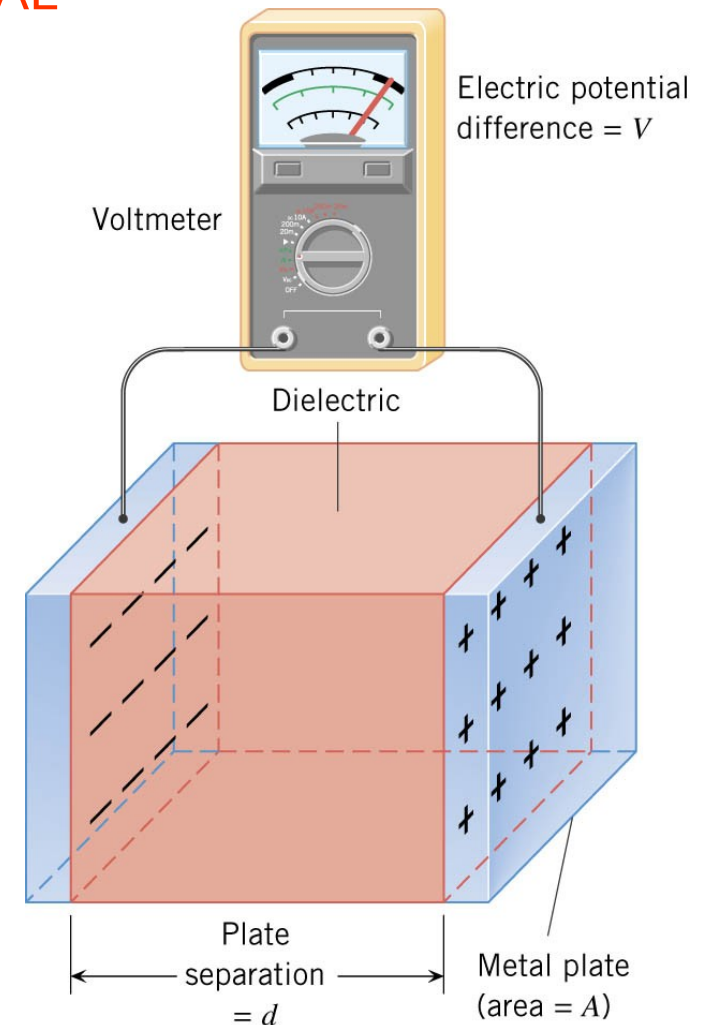
The magnitude of the charge in each plate of the capacitor is directly proportional to the magnitude of the potential difference between the plates.

$$q = CV \quad \text{Or } V = q / C$$

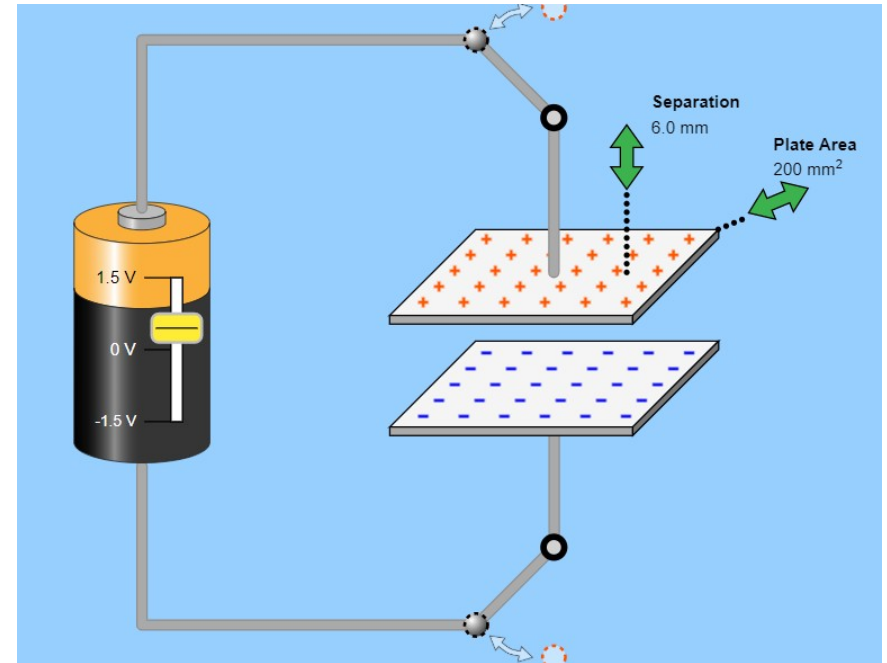
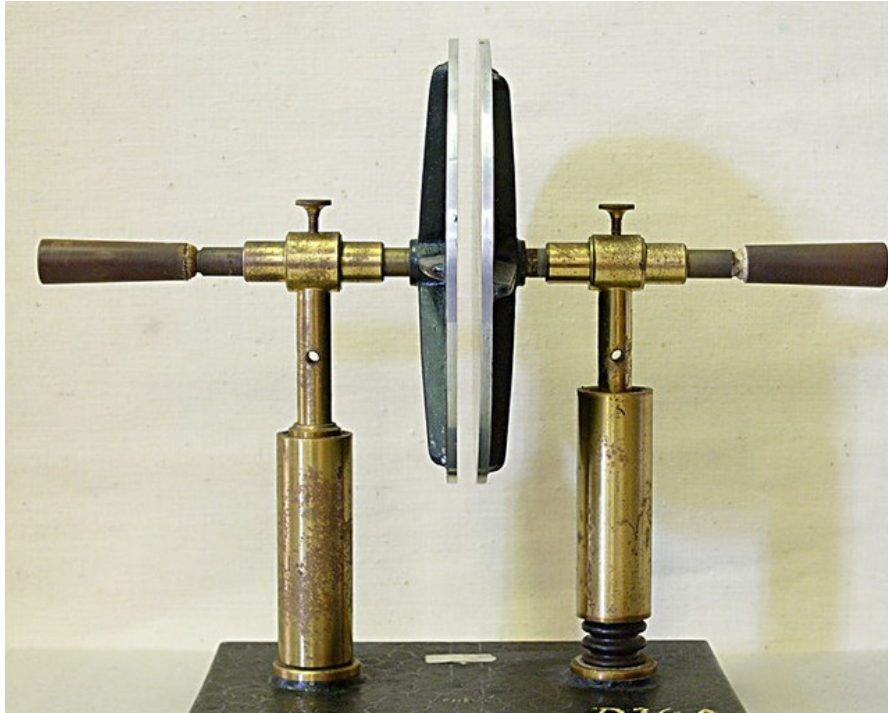
The capacitance C is the proportionality constant.

SI Unit of Capacitance: coulomb/volt = farad (F)

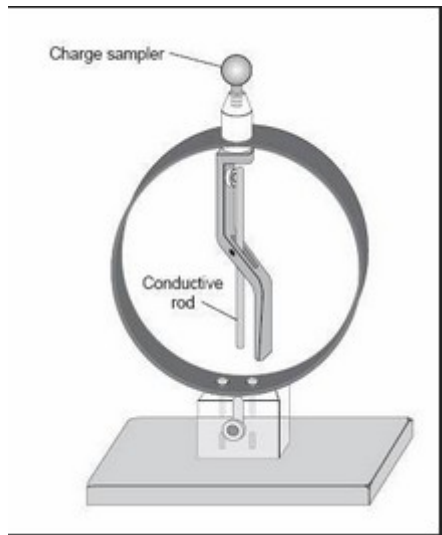
Think of capacity. The capacitance is the charge Per unit volt. So if you charge a 4pF with a 9V Battery then the maximum charge is ? _____
Pico is 10^{-12}



Special case → most common cap → plate capacitor.



https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html



The capacitance is proportional to area
And inversely proportional to distance between the plates.

Special case: plate capacitor

Capacitance Depends on the distance between the plates and the size of the plates

Equation

$$C = \frac{\epsilon_0 A}{d}$$

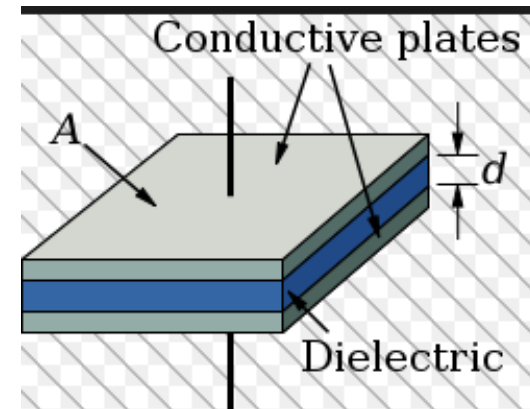
Capacitance $\rightarrow$ C

Electrostatic constant $\rightarrow$ ϵ_0

Area $\rightarrow$ A

Distance Between plates $\rightarrow$ d

$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$

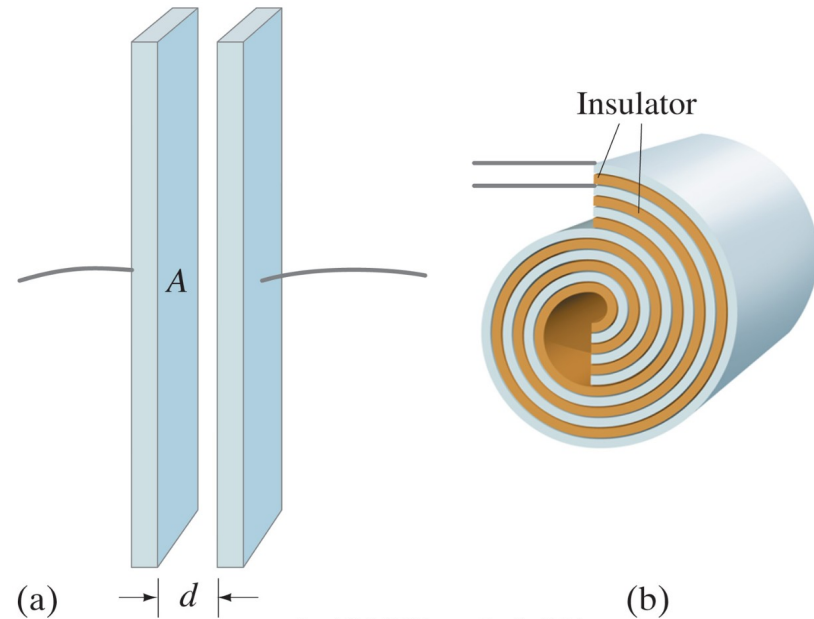


$$1\text{mm} = 10^{-3} \text{ m}$$

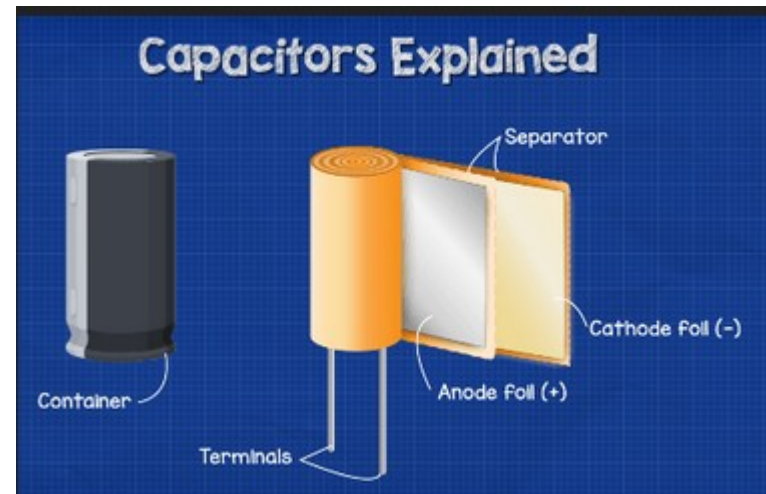
The dielectric increases the capacitance by a factor k the dielectric constant.

(a) What is the capacitance of a parallel plate capacitor with metal plates, each of area 1.00 m^2 , separated by 1.00 mm ? (b) What charge is stored in this capacitor if a voltage of $3.00 \times 10^3 \text{ V}$ is applied to it?

<https://phet.colorado.edu/en/simulation/legacy/capacitor-lab>



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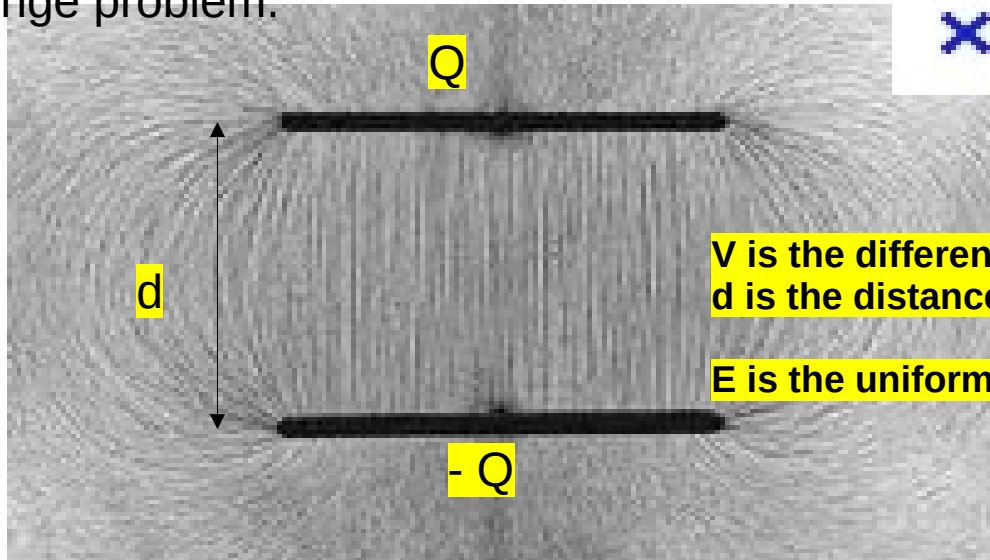
<http://www.ladyada.net/make/sudc4kap/make.html>

$$Q = C V \quad \text{and} \quad E = V/d$$

$$\text{Energy stored} = 0.5 Q V = 0.5 C V^2$$

Show the simulation and see if equations work

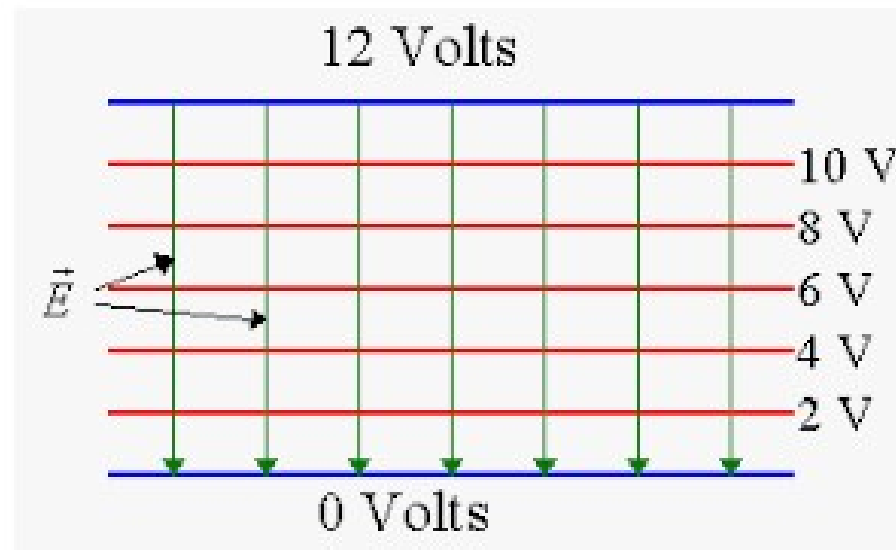
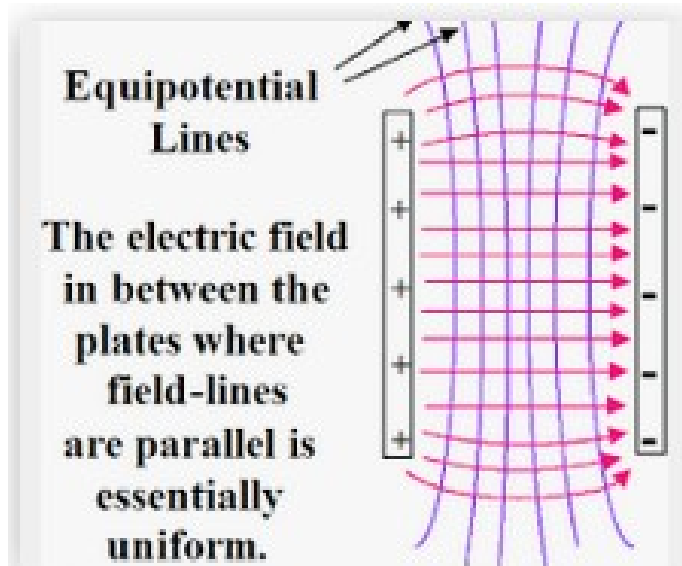
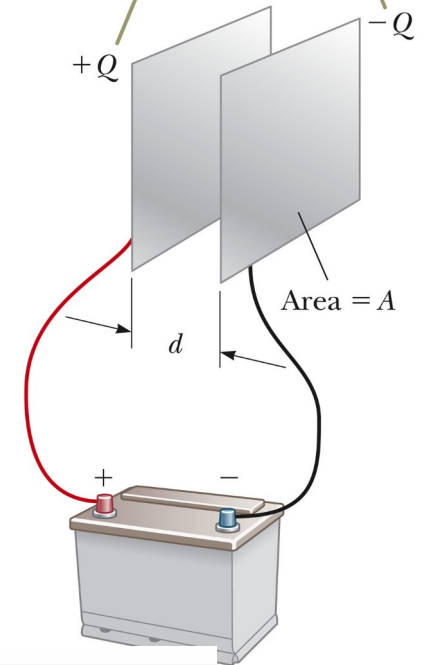
Example: parallel plate capacitor. Electric field is homogeneous between plates. Fringe problem.



V is the difference in potential or voltage.
 d is the distance between the plates

E is the uniform electric field from + to -

When the capacitor is connected to the terminals of a battery, electrons transfer between the plates and the wires so that the plates become charged.



<https://phet.colorado.edu/en/simulation/capacitor-lab>

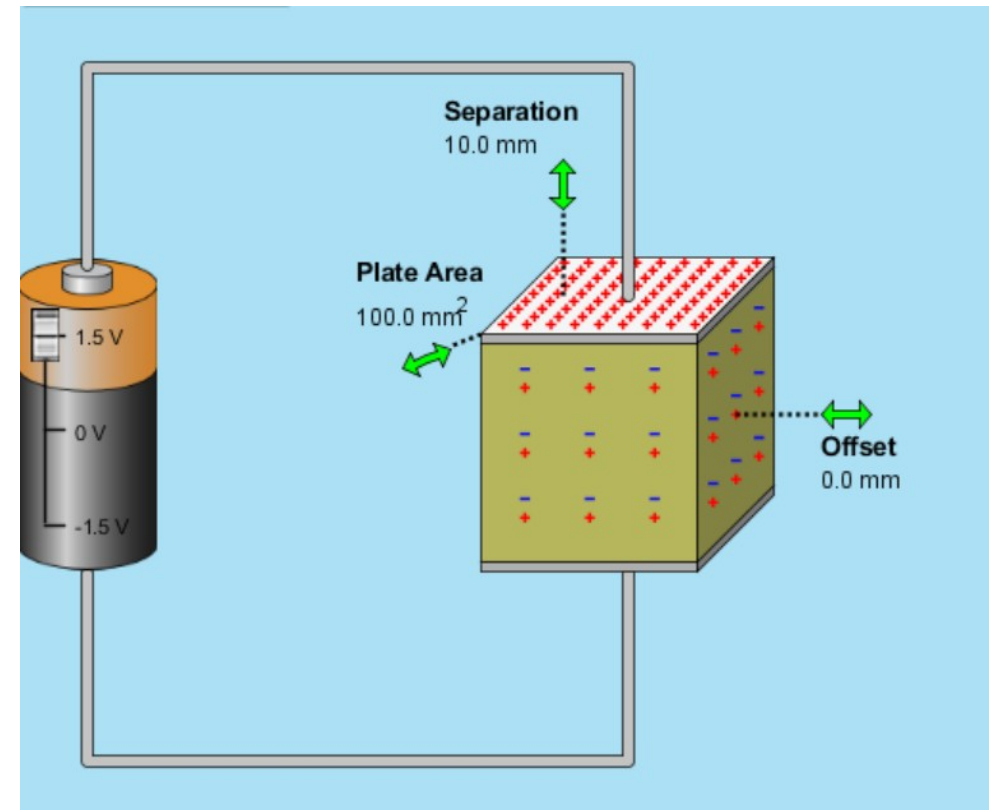
A parallel plate capacitor consists of two metal plates, one carrying charge $+q$ and the other carrying charge $-q$.

It is common to fill the region between the plates with an electrically insulating substance called a **dielectric**.

$$C = K C_0$$

$$C = Q/V$$

$$V = E d$$



<https://phet.colorado.edu/sims/cheerpj/capacitor-lab/latest/capacitor-lab.html?simulation=capacitor-lab>

Mica is 7

Material	Dielectric constant κ
Vacuum	1.00000
Air	1.00059
Bakelite	4.9
Fused quartz	3.78
Neoprene rubber	6.7
Nylon	3.4
Paper	3.7
Polystyrene	2.56
Pyrex glass	5.6
Silicon oil	2.5
Strontium titanate	233
Teflon	2.1
Water	80



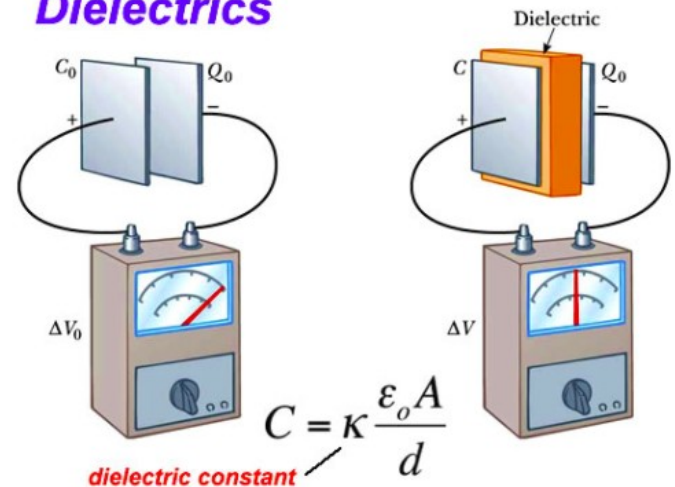
See simulation. The dielectric increases the capacitance.

$$C = \kappa \epsilon_0 \times A/d$$

$$\epsilon_0 = 8.854 \times 10^{-12}$$

κ is the dielectric constant.

Dielectrics



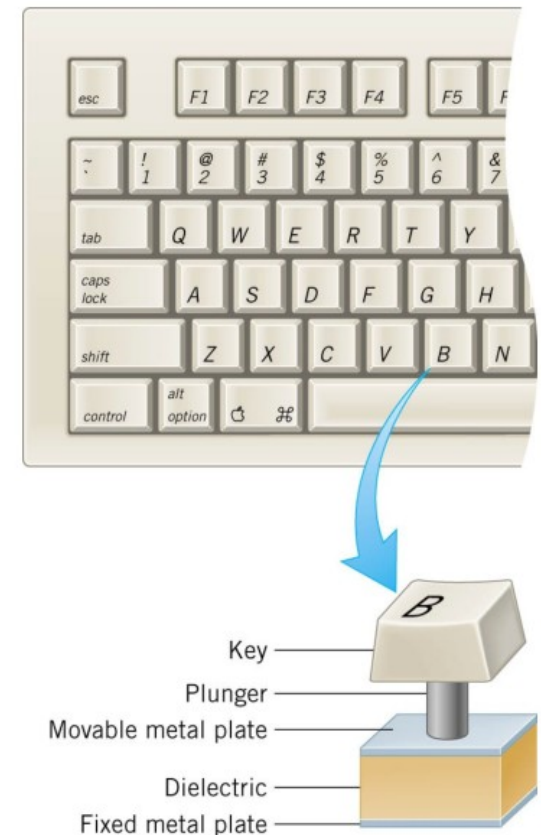
https://www.electronics-tutorials.ws/capacitor/cap_2.html

Example 12 A Computer Keyboard

One common kind of computer keyboard is based on the idea of capacitance. Each key is mounted on one end of a plunger, the other end being attached to a movable metal plate. The movable plate and the fixed plate form a capacitor. When the key is pressed, the capacitance increases. The *change* in capacitance is detected, thereby recognizing the key which has been pressed.

The separation between the plates is 5.00 mm, but is reduced to 0.150 mm when a key is pressed. The plate area is $9.50 \times 10^{-5} \text{ m}^2$ and the capacitor is filled with a material whose dielectric constant is 3.50.

Determine the change in capacitance detected by the computer.

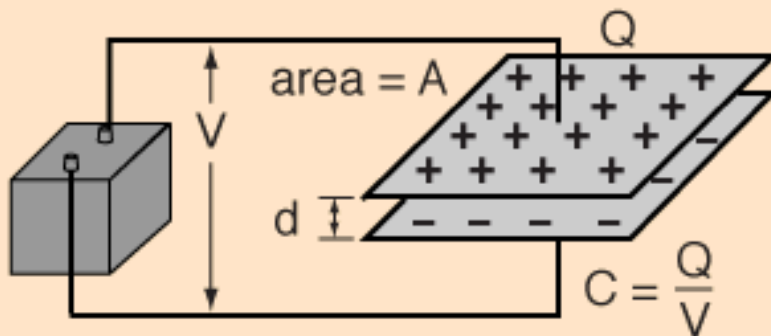


$$C = \frac{\kappa \epsilon_o A}{d} = \frac{(3.50)(8.85 \times 10^{-12} \text{ C}^2 / (\text{N} \cdot \text{m}^2))(9.50 \times 10^{-5} \text{ m}^2)}{0.150 \times 10^{-3} \text{ m}} = 19.6 \times 10^{-12} \text{ F}$$

$$C = \frac{\kappa \epsilon_o A}{d} = \frac{(3.50)(8.85 \times 10^{-12} \text{ C}^2 / (\text{N} \cdot \text{m}^2))(9.50 \times 10^{-5} \text{ m}^2)}{5.00 \times 10^{-3} \text{ m}} = 0.589 \times 10^{-12} \text{ F}$$

$$\Delta C = 19.0 \times 10^{-12} \text{ F}$$

Electric Field Energy in Capacitor



The energy stored on a capacitor is in the form of energy density in an electric field is given by

$$\eta_E = \frac{\text{energy}}{\text{volume}} = \frac{1}{2} \epsilon E^2$$

This can be shown to be consistent with the energy stored in a charged parallel plate capacitor

$C_{\text{parallel plate}} = k\epsilon_0 \frac{A}{d}$	Show
---	----------------------

Capacitor energy = $\frac{1}{2} CV^2$	Show
---------------------------------------	----------------------

Electric field = $E = \frac{V}{d}$	Show
------------------------------------	----------------------

$$\begin{aligned} \text{Energy} &= \eta V = \frac{1}{2} \epsilon E^2 A d \\ &= \frac{1}{2} \epsilon \frac{V^2}{d^2} A d \\ &= \frac{1}{2} \frac{\epsilon A}{d} V^2 = \frac{1}{2} CV^2 \end{aligned}$$

($\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$). The dielectric constant (k) of a material is the ratio of its permittivity ϵ to the permittivity of vacuum ϵ_0 , so $k = \epsilon / \epsilon_0$.

What charge is stored in a $180\ \mu\text{F}$ capacitor when $120\ \text{V}$ is applied to it?

What is the electric field between the plates? How much energy is it stored inside?

(a) What is the capacitance of a parallel plate capacitor with metal plates, each of area $1.00\ \text{m}^2$, separated by $1.00\ \text{mm}$? (b) What charge is stored in this capacitor if a voltage of $3.00 \times 10^3\ \text{V}$ is applied to it?

Find the charge stored when $5.50\ \text{V}$ is applied to an $8.00\ \text{pF}$ capacitor.

Calculate the voltage applied to a $2.00\ \mu\text{F}$ capacitor when it holds $3.10\ \mu\text{C}$ of charge.

What capacitance is needed to store $3.00\ \mu\text{C}$ of charge at a voltage of $120\ \text{V}$?

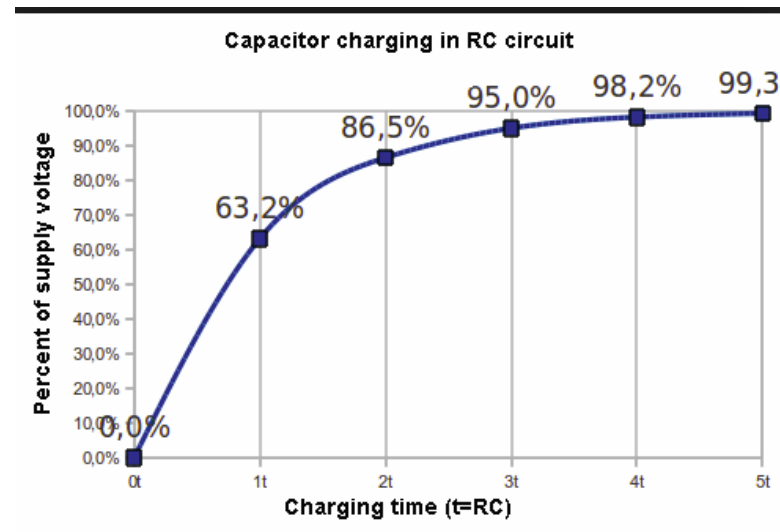
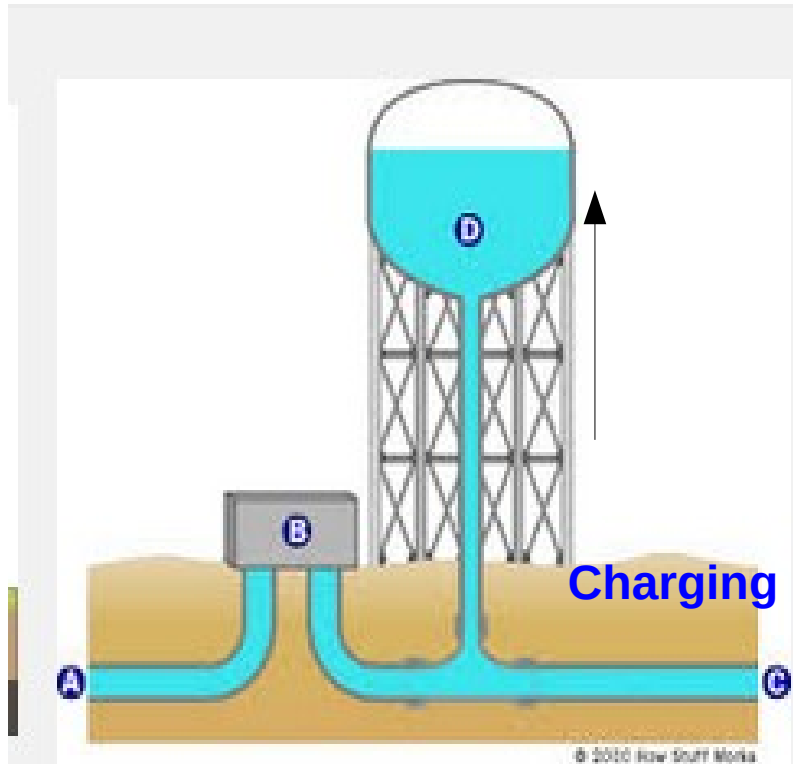
What is the capacitance of a large Van de Graaff generator's terminal, given that it stores $8.00\ \text{mC}$ of charge at a voltage of $12.0\ \text{MV}$?

Find the capacitance of a parallel plate capacitor having plates of area $5.00\ \text{m}^2$ that are separated by $0.100\ \text{mm}$ of Teflon.

(a) What is the capacitance of a parallel plate capacitor having plates of area $1.50\ \text{m}^2$ that are separated by $0.0200\ \text{mm}$ of neoprene rubber? (b) What charge does it hold when $9.00\ \text{V}$ is applied to it?

Charging → voltage across the cap as a function of time

Note: charge follows the same graph ($Q = C V$) – the current decreases.

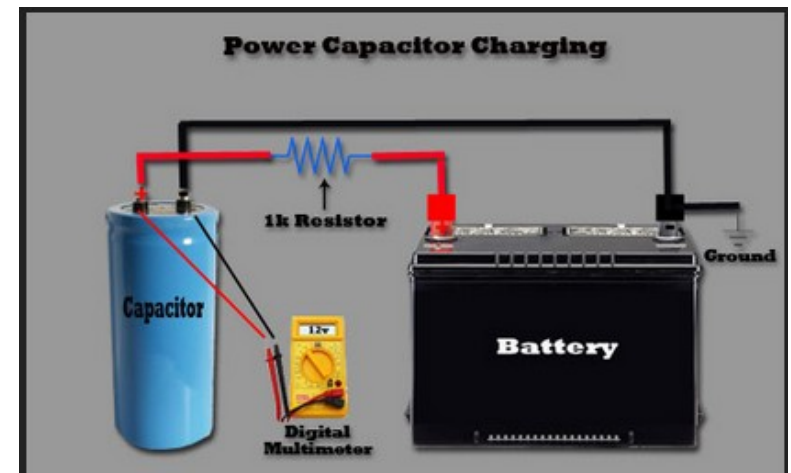


Charging → → At the beginning it is easy to pump the water up
Then it gets harder because of the weight of
The water. Same idea when you charge a capacitor

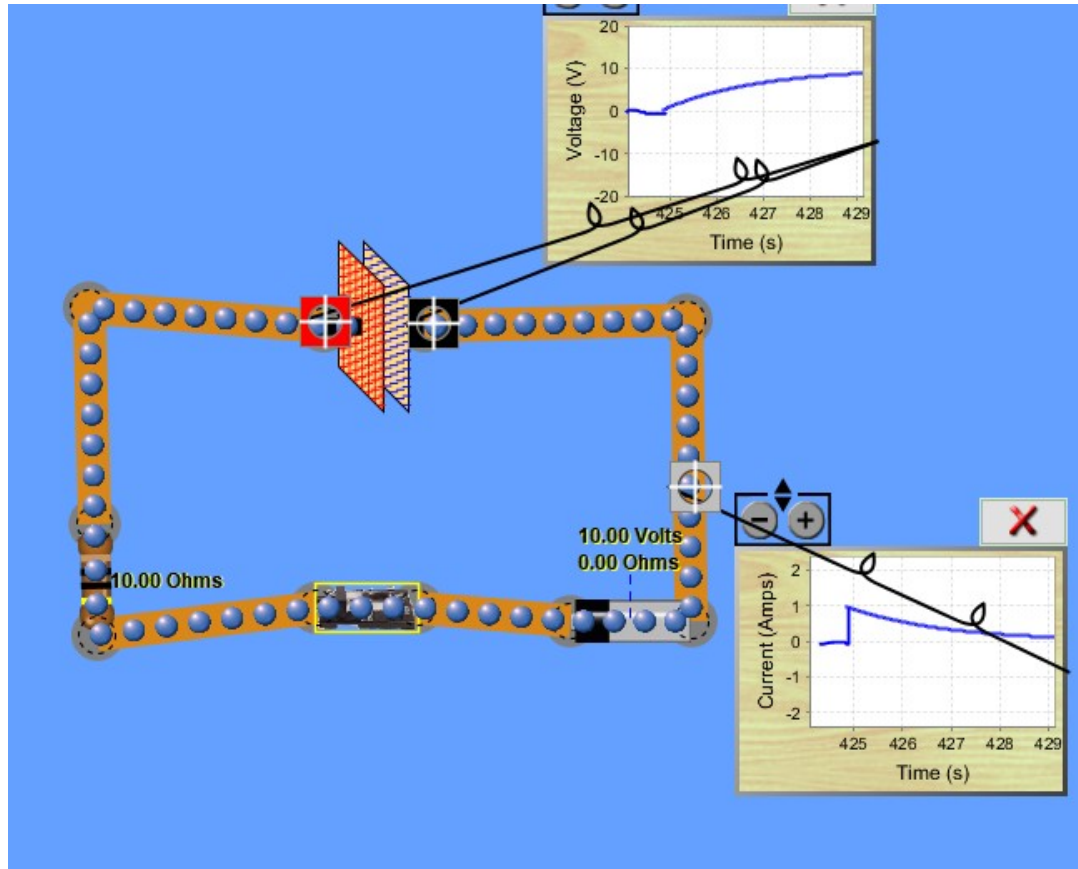
$T = RC$ is called the time constant of the capacitor.
It is the time it takes to charge it at 63%.

Do exploration of Physics app (RC)

Do some Math.



Charging → voltage across the cap as a function of time



This is charging a capacitor
Through a resistance

The voltage of the battery is 10V
The resistance is 10.00 ohms
The capacitance is 0.2 Farads

The time constant is $RC = 2s$
It is the time to get to 63% of the
Maximum voltage 10V.

Observe the voltage across the
Capacitor. It is increasing until
It reaches 10V.

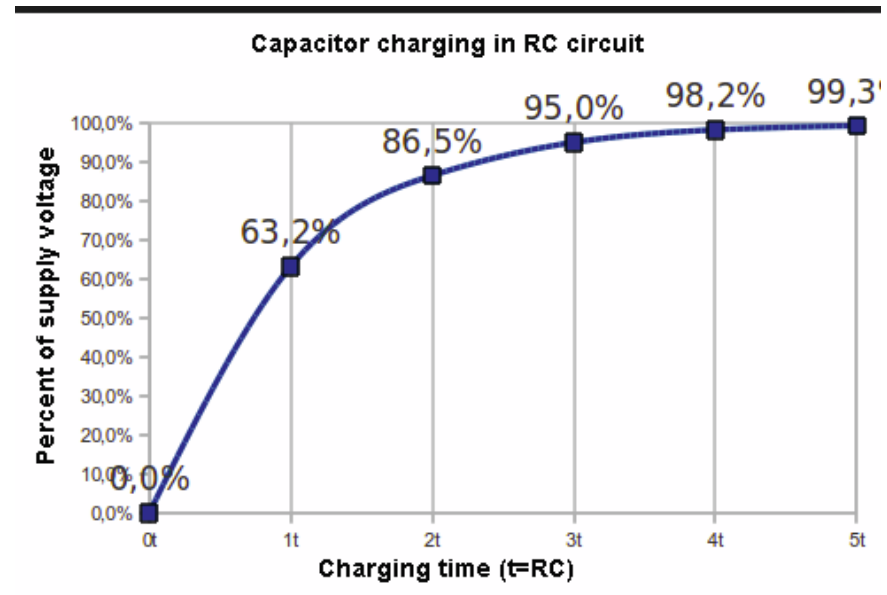
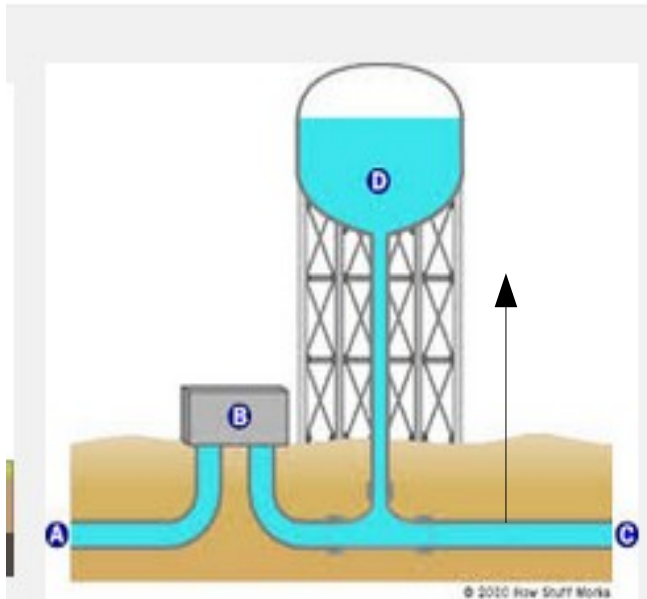
What is 63% of the max value ?
Check if after $t = RC$ the voltage is
Indeed about 63% of 50V

This graph shows the voltage across the capacitor
as a function of time. $V(t)$. The voltage increases from
0 to 10V. First the voltage increases quickly then
Slowly. The slope is steep then flats out.

Note for me: capacitor_app.jar from phys116

[https://phet.colorado.edu/en/simulation/
circuit-construction-kit-ac](https://phet.colorado.edu/en/simulation/circuit-construction-kit-ac)

discharging → voltage across the cap as a function of time



<http://micro.magnet.fsu.edu/electromag/java/capacitor/index.html>

Equation of charge **$V(t) = V_{\max} (1 - e^{-(t/RC)})$**

$V_{\max}$ is the voltage of power supply used

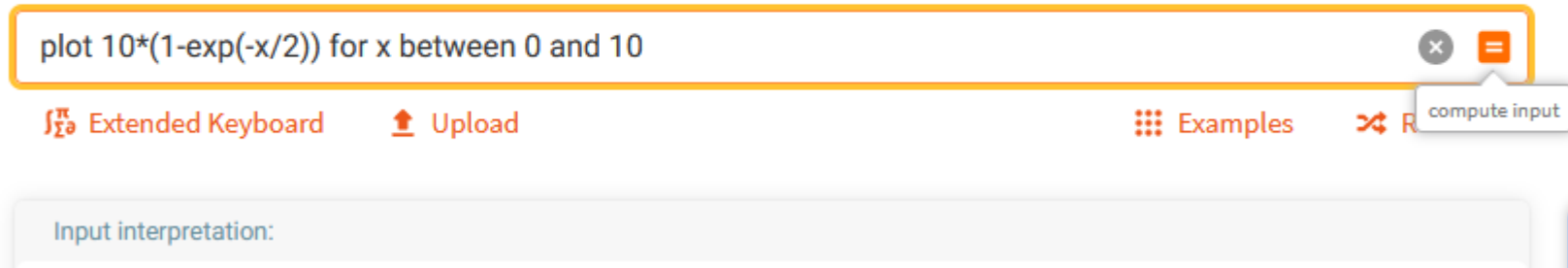
To charge the capacitor.

R is the resistance through which the cap is charging

C is the capacitance of the capacitor.

$V(t)$ is the voltage across the capacitor.

Suppose you are charging a capacitor through a resistor of 10 ohms.
The capacitance is 0.2F . What is the equation?



1) Suppose you are charging a capacitor through a resistor of 10 ohms. The capacitance is 0.2F . What is the equation?

The equation of the charge is:

$$V(t)=10(1-e^{-t/2})$$

Plug that in wolfram alpha.

Check that when $t=2s$

The voltage is about 63% of 10

Use the equation to find the voltage when

$t=1s, 3s, 4s, 5s$ (use your TI then check with the graph

2) Suppose you are charging a capacitor with a power supply of 20V

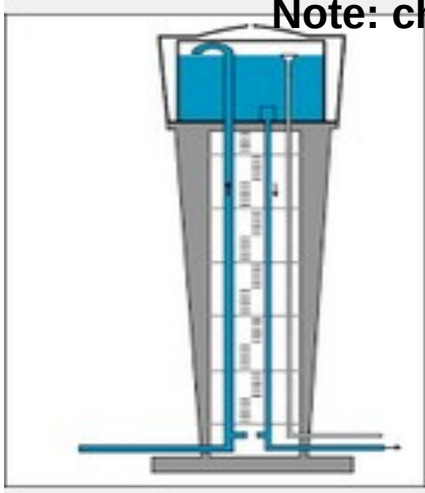
The capacitance is 0.1F and the resistance is 100 ohms

What is RC ? What is the voltage across capacitor when $t = RC$

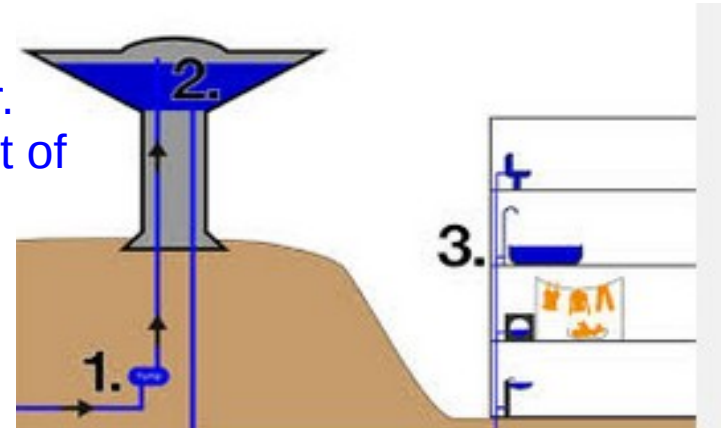
What is the equation $V(t)$? What is the voltage when $t = 5s, 10s, 20s, 30s$? Use TI)

discharging → voltage across the cap as a function of time

Note: charge follows same pattern



Like emptying the tower.
First it's fast. The weight of
The water is pushing
Then it slows down.

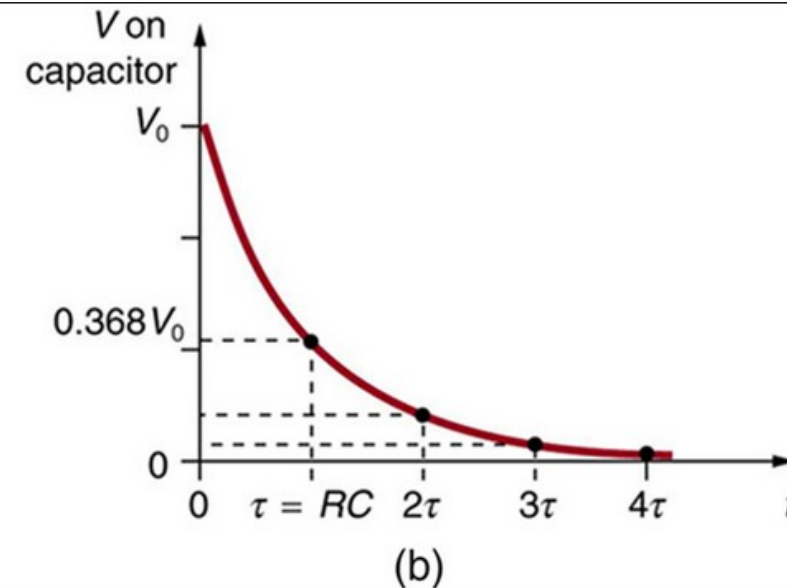
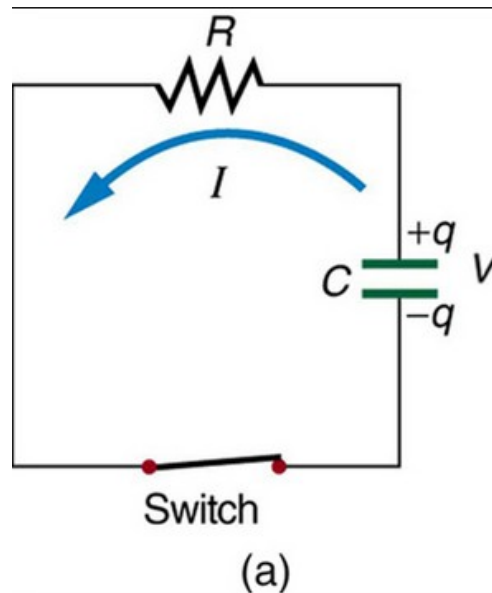


<http://micro.magnet.fsu.edu/electromag/java/capacitor/index.html>

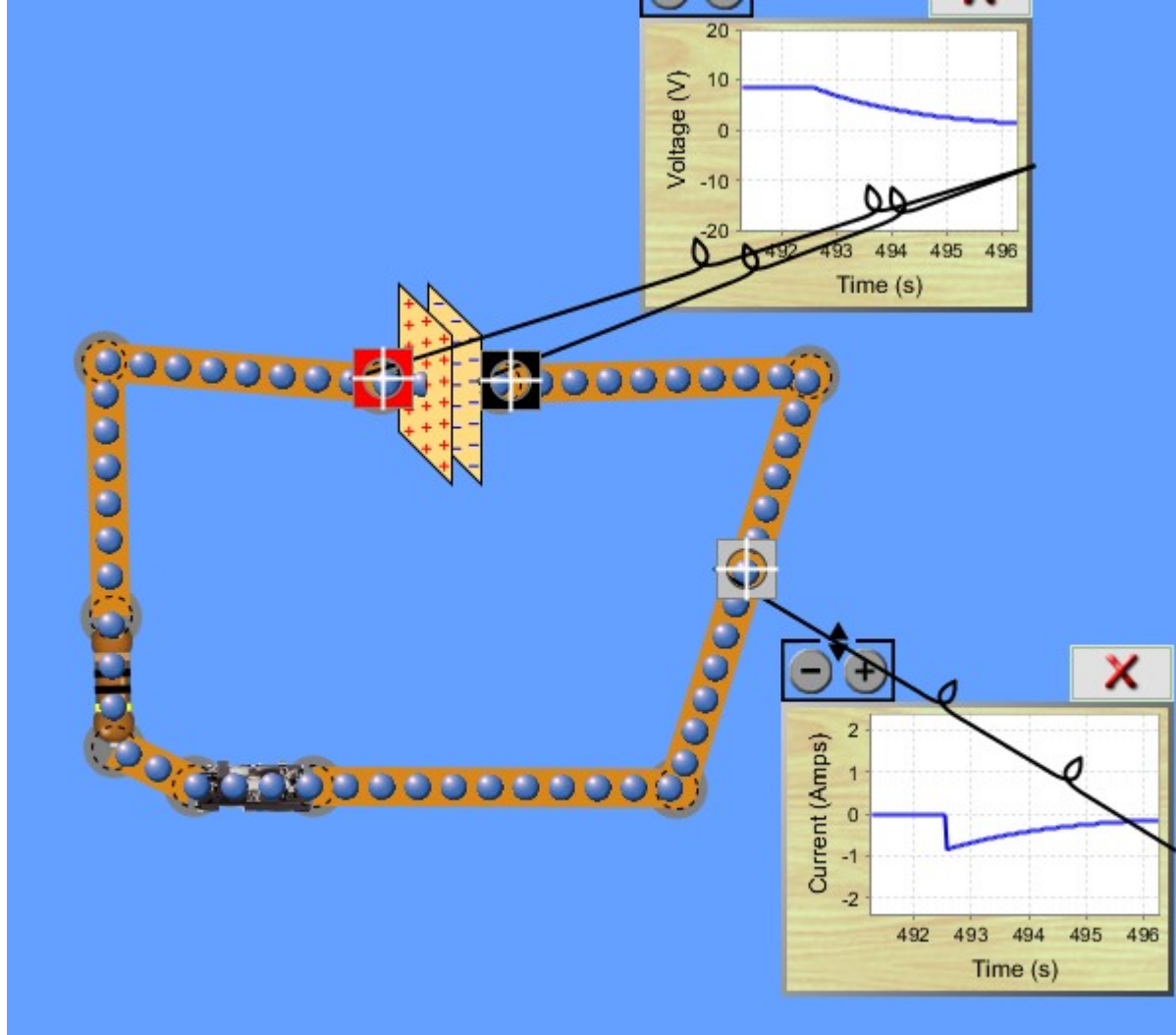
RC is the time for the
Capacitor to lose
63% of its charge
(or voltage)
It remains 37% of the
initial voltage.

It is an exponential decay

$$V = V_{\max} e^{(-t/RC)}$$



Discharging in a load



ohms

plot $10 \cdot (\exp(-x/2))$ for x between 0 and 10

Extended Keyboard Upload

Input interpretation:

plot	$10 \exp\left(-\frac{x}{2}\right)$	$x = 0 \text{ to } 10$
------	------------------------------------	------------------------

Use wolfram alpha to plot the exponential

With the TI, find V for $t = 0, 1, 2, 3..$
Check with the line

A 10V voltage is used to charge a capacitor through a 500 ohms resistor.

The capacitance of the capacitor is 0.01 F (micro farad)

A) What is the time constant in seconds ?

B) The capacitor is now discharging in the resistance.

C) What is the equation of the voltage across the capacitor ?

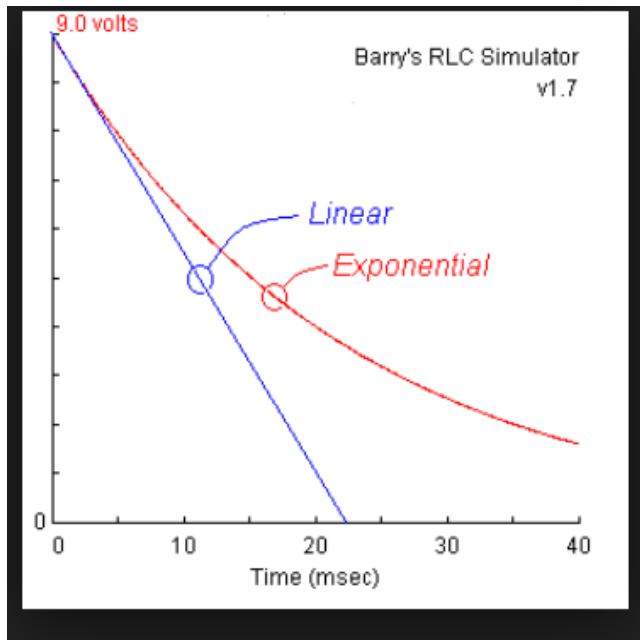
D) Plot the equation for voltage for x between 0 and 10

E) Use the equation to find the voltage at 2 s , 4 s , 6s, 8s check with graph,

F) The time constant is the time it takes for voltage to decrease by 37%. 37% of 10 is _____ -

Use your graph to find the time it takes for the voltage to decrease by 37%

Compare to the time constant $R * C$



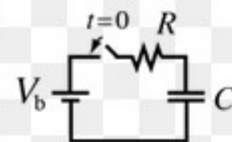
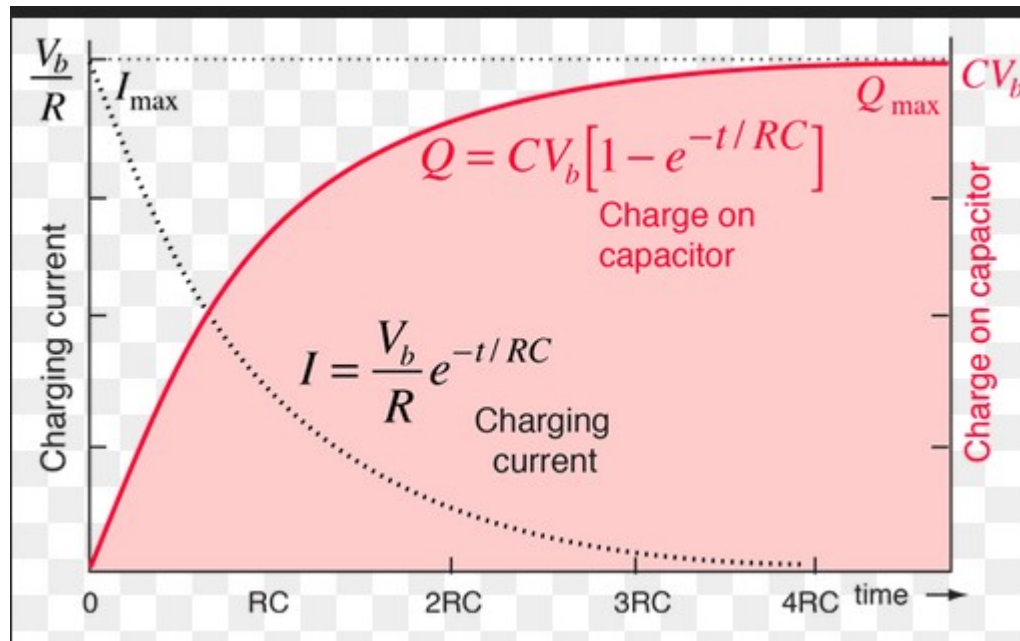
$$V = V_0 e^{-\frac{t}{RC}}$$

$$\ln V = \ln(V_0) - t / RC$$

$1/RC$ is the slope of the line.

Using oscilloscope:





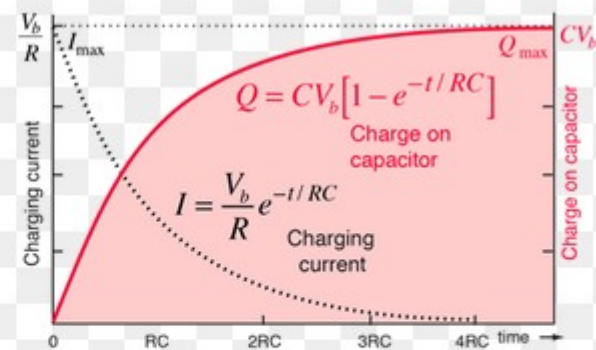
$$V_b = V_R + V_C$$

$$V_b = IR + \frac{Q}{C}$$

As charging progresses,

$$V_b = IR + \frac{Q}{C}$$

$\downarrow$ $\uparrow$
 current decreases and charge increases.



At $t = 0$

$$Q = 0$$

$$V_C = 0$$

$$I = \frac{V_b}{R}$$

As $t \rightarrow \infty$

$$Q \rightarrow CV_b$$

$$V_C \rightarrow V_b$$

$$I \rightarrow 0$$

Vectors	Force (F)	$F_e = k_c \frac{q_1 q_2}{r^2}$ (in N)
	Field (potential for a force)	$E = k_c \frac{q_1}{r^2}$ (in N/C)

F force (N) between 2 charges- it is a vector
 r is the distance between the charges
 K = 9×10^9 (constant)
 Q1 and q2 are the charges (Coulombs_

E electric field (N/C) generated by a charge q at a distance r
 From the charge
 K = 9×10^9 q is the charge (C)

F = q E A force q placed in an electric field E will experience a force F

Potential (for energy) or Voltage	$V = k_c \frac{q_1}{r}$ (in J/C)
--------------------------------------	-------------------------------------

V Potential (volts) at a distance r from a charge q1
 It is the energy of a 1C charge placed at a distance r from q

C = ϵ_0 A/d capacitance (F) of a plate capacitor with air between the plates
 $\epsilon_0 = 8.854 \times 10^{-12}$ F/m
 A is the area in m^2
 .d is the distance between the plates in meters

C = k ϵ_0 x A/d capacitance of a plate capacitor with a dielectric of constant k (k=1 if there is air between)

Q = C V charge (coulombs) stored in a plate capacitor
 V is the voltage between the plates
 C is the capacitance in Farad

E=V/d electric field inside a plate capacitor in N/C
 V is voltage in volts (across the capacitor)
 d is distance in m between the plate

Energy stored in a plate capacitor: **U = 0.5 Q V = 0.5 C V²**